INTERPLM: DISCOVERING INTERPRETABLE FEATURES IN PROTEIN LANGUAGE MODELS VIA SPARSE AUTOENCODERS

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ABSTRACT

Protein language models (PLMs) have demonstrated remarkable success in protein modeling and design, yet their internal mechanisms for predicting structure and function remain poorly understood. Here we present a systematic approach to extract and analyze interpretable features from PLMs using sparse autoencoders (SAEs). By training SAEs on embeddings from the PLM ESM-2, we identify up to 2,548 human-interpretable latent features per layer that strongly correlate with up to 143 known biological concepts such as binding sites, structural motifs, and functional domains. In contrast, examining individual neurons in ESM-2 reveals up to 46 neurons per layer with clear conceptual alignment across 15 known concepts, suggesting that PLMs represent most concepts in superposition. Beyond capturing known annotations, we show that ESM-2 learns coherent concepts that do not map onto existing annotations and propose a pipeline using language models to automatically interpret novel latent features learned by the SAEs. As practical applications, we demonstrate how these latent features can fill in missing annotations in protein databases and enable targeted steering of protein sequence generation. Our results demonstrate that PLMs encode rich, interpretable representations of protein biology and we propose a systematic framework to extract and analyze these latent features. In the process, we recover both known biology and potentially new protein motifs. As community resources, we introduce InterPLM (interPLM.ai), an interactive visualization platform for exploring and analyzing learned PLM features, and release code for training and analysis at github.com/ElanaPearl/interPLM.

***K*eywords** Protein Language Models · Mechanistic Interpretability · Machine Learning · Structural Biology

1 Introduction

Language models of protein sequences have revolutionized our ability to predict protein structure and function [1, 2]. These models achieve their impressive performance by learning rich representations from the vast diversity of naturally evolved proteins. Their success raises a fundamental question: what exactly do protein language models (PLMs) learn? Advances in interpretability research methods now enable us to investigate this question and understand at least some of the representations they have learned in service of this challenging task.

Gaining insights into the internal mechanisms of PLMs is crucial for model development and biological discovery [3]. Understanding how these models process information enables more informed decisions about model design, beyond merely optimizing performance metrics. By analyzing the features driving predictions, we can identify spurious correlations or biases, assess the generalizability of learned representations, and potentially uncover novel biological insights. For instance, models might learn to predict certain protein properties based on subtle patterns or principles that have eluded human analysis, opening up new avenues for experimental investigation. Conversely, detecting biologically implausible features can guide improvements to the models' inductive biases and learning algorithms. This systematic analysis of prediction mechanisms offers opportunities to enhance model performance and reliability while extracting biological hypotheses from learned representations. Furthermore, PLM interpretation can illuminate how much these models learn genuine physical and chemical principles governing protein structure, as opposed to simply memorizing structural motifs. As PLMs continue to advance, systematic interpretation frameworks may uncover increasingly sophisticated biological insights.

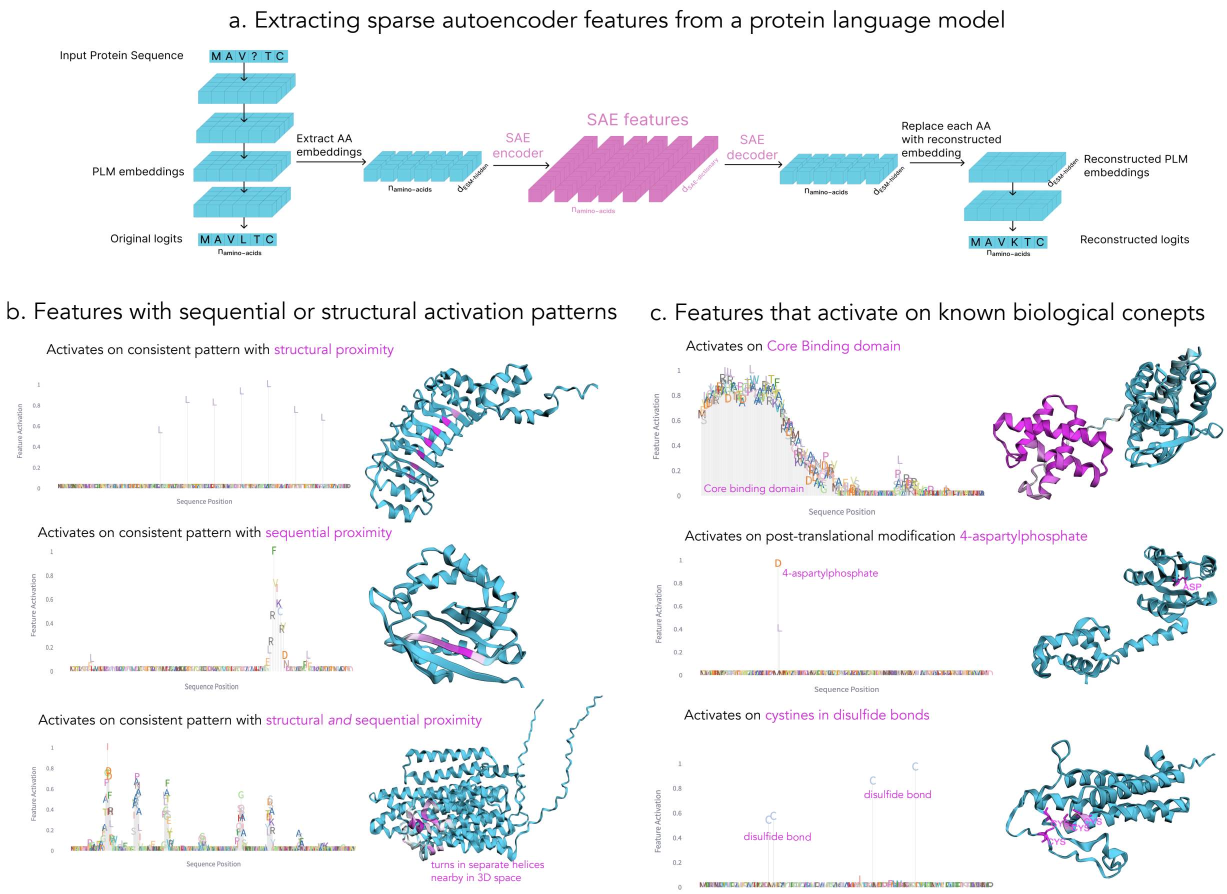

Figure 1: **Overview of SAE methodology and representative SAE features revealed through automated activation pattern analysis.** a) Pipeline illustrating the extraction of embeddings, their conversion to features, and subsequent reinsertion of reconstructed embeddings into the PLM. b-c) Examples of features exhibiting interpretable activation patterns, both structural and conceptual. Each feature is visualized using a protein where maximal activation occurs. Feature activation intensities are displayed both along the protein sequence (amino acid height indicates activation magnitude) and protein structure (darker pink indicates stronger activation). b) Features selected to demonstrate activation patterns in structurally proximate amino acids, sequentially proximate amino acids, or both. c) Features selected based on significant associations between activation patterns and known Swiss-Prot biological concept annotations. Feature identifiers and corresponding layers (top to bottom): Left panel (f/3147, f/10091, f/67, layer 4); Right panel (f/7125, f/8128, f/1455 from layers 4, 5, 5).

In this work, we create a versatile framework using sparse auto-encoders (SAEs) to interpret latent features learned by PLMs. To demonstrate this framework, we analyze amino acid embeddings across all layers of ESM-2 [1]. Using several scalable methods we developed for identifying and annotating protein SAE features, we discovered that ESM-2's latent features accurately capture many known biophysical properties - including catalytic active sites, zinc finger domains, mitochondrial targeting sequences, phosphorylated residues, disulfide bonds, and disordered regions. By quantitatively comparing these identified features to known concept annotations, we establish new evaluation metrics for SAEs. We further use Claude-3.5 Sonnet (new) [4] to provide automated annotations of latent features and use PLM feature activation patterns to identify missing and potentially new protein annotations. We show how PLM features can be used to steer model outputs in interpretable ways. To facilitate exploration of these features, we provide an interactive visualization platform `interPLM.ai`, along with code for training and analyzing features at `github.com/ElanaPearl/interPLM`.

2 Related Works & Background

2.1 PLM Interpretability

PLMs are deep learning models (typically transformers) that perform self-supervised training on protein sequences, treating amino acids as tokens in a biological language to learn their underlying relationships and patterns [5]. These models consist of multiple transformer layers, each containing attention mechanisms and neural networks that progressively build up intermediate representations of the protein sequence. Prior work on interpreting PLMs has focused on analyzing these internal representations - both by probing the hidden states at different layers and by examining the patterns of attention between amino acids. Studies have demonstrated that attention maps can reveal protein contacts [6] and binding pockets, while the hidden state representations from different layers can be probed to predict structural properties like secondary structure states [7]. Recent evidence suggests that rather than learning fundamental protein physics, PLMs primarily learn and store coevolutionary patterns - coupled sequence patterns preserved through evolution [3]. This finding aligns with traditional approaches that explicitly modeled coevolutionary statistics [8], and with modern deep learning methods that leverage evolutionary relationships in training [9].

However, even if PLMs are memorizing evolutionarily conserved patterns in motifs, key questions remain about their internal mechanisms: How do they identify conserved motifs from individual sequences? What percentage of learned features actually focus on these conserved motif patterns? How do they leverage these memorized patterns for accurate sequence predictions? What additional computational strategies support these predictions? Answering these questions could both reveal valuable biological insights and guide future model development.

2.2 Sparse Autoencoders (SAEs)

In attempts to reverse-engineer neural networks, researchers often analyze individual neurons - the basic computational units that each output a single activation value in response to input. However, work in mechanistic interpretability has shown that these neurons don't map cleanly to individual concepts, but instead exhibit superposition - where multiple unrelated concepts are encoded by the same neurons [10, 11]. Sparse Autoencoders (SAEs) are a dictionary learning approach that addresses this by transforming each neuron's activation into a larger but sparse hidden layer [12, 13, 14].

At their core, SAEs learn a "dictionary" of sparsely activated features that can reconstruct the original neuron activations. Each feature i is characterized by two components: its dictionary vector ($\mathbf{d}_i$) in the original embedding space (stored as rows in the decoder matrix) and its activation value (f_i) that determines its contribution. The reconstruction of an input activation vector $\mathbf{x}$ can be expressed as:

$$\mathbf{x} \approx \mathbf{b} + \sum_{i=1}^{d_{\text{dict}}} f_i(\mathbf{x})\mathbf{d}_i$$

where $\mathbf{b}$ is a bias term and d_{dict} is the size of the learned dictionary. This decomposition allows us to represent complex neuron activations as combinations of more interpretable features (see Appendix Figure 8 for more details).

SAE analysis has advanced our understanding of how language and vision models process information [13, 15]. Neural network behavior can be understood through computational circuits - interconnected neurons that collectively perform specific functions. While traditional circuit analysis uncovers these functional components (like edge detectors [11] or word-copying mechanisms [16]), using SAE features instead of raw neurons has improved the identification of circuits responsible for complex behaviors [17].

Researchers can characterize these features through multiple approaches: visual analysis [18], manual inspection [14], and large language model assistance [19]. Feature functionality can be verified through intervention studies, where